Vortex Panel Method

1. Panel geometry

$$x_i = \tfrac{1}{2}\left(X_i + X_{i+1}\right), \qquad y_i = \tfrac{1}{2}\left(Y_i + Y_{i+1}\right) \tag{1}$$

$$S_j = \sqrt{\Delta X_j^2 + \Delta Y_j^2} \tag{2}$$

$$\hat{\mathbf{n}}_i = \frac{1}{S_i}\left(-\Delta Y_i,\ \Delta X_i\right) \tag{3}$$

2. Influence of panel j on control point i

$$\theta_{ij} = \tan^{-1}\left(\frac{y_i - y_j}{x_i - x_j}\right) \tag{4}$$

$$\frac{\partial \theta_{ij}}{\partial n_i} = \frac{-(y_i - y_j)\, n_{x,i} + (x_i - x_j)\, n_{y,i}}{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{5}$$

$$J_{ij} = \int_j \frac{\partial \theta_{ij}}{\partial n_i}\, ds_j \ \approx \ \begin{cases} \dfrac{\partial \theta_{ij}}{\partial n_i}\, S_j, & i \neq j \\ 0, & i = j \end{cases} \tag{6}$$

3. Flow tangency at each control point

$$V_\infty \cos\beta_i = V_\infty \left(\cos\alpha\, n_{x,i} + \sin\alpha\, n_{y,i}\right) \tag{7}$$

$$V_\infty \cos\beta_i - \sum_{j=1}^{n} \frac{\gamma_j}{2\pi}\, J_{ij} = 0, \qquad i = 1, \ldots, n \tag{8}$$

4. Kutta condition

$$\gamma_1 + \gamma_n = 0 \tag{9}$$

This replaces the tangency equation at one panel ($i = n/2$), so there are n equations for n unknowns.

5. Lift

$$\Gamma = \sum_{j=1}^{n} \gamma_j S_j \tag{10}$$

$$L' = \rho V_\infty \Gamma \tag{11}$$

$$c_l = \frac{L'}{\frac{1}{2}\rho V_\infty^2 c} = \frac{2\Gamma}{V_\infty c} \tag{12}$$

6. Surface pressure

$$C_{p,i} = 1 - \left(\frac{\gamma_i}{V_\infty}\right)^2 \tag{13}$$